

PORTFOLIO CASE STUDY • 2023 TO 2026

Hangman: Rescue Mission

A bootcamp word game rebuilt as a production app on Cloudflare's free tier, at a running cost of \$0.



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A bootcamp game with one flaw I could not keep

2023 • MERN bootcamp

Built at Social Hackers Academy with React, an Express server, MongoDB and OpenAI.

A game for learners

Guess the word to learn English vocabulary and pronunciation. Win a round and a painting rescues the robot.

What was wrong

Example sentences were LLM-written quotations attributed to real authors and works. That was wrong.

Rebuilt, not migrated.

Real content only

Examples come from Merriam-Webster, with attribution.

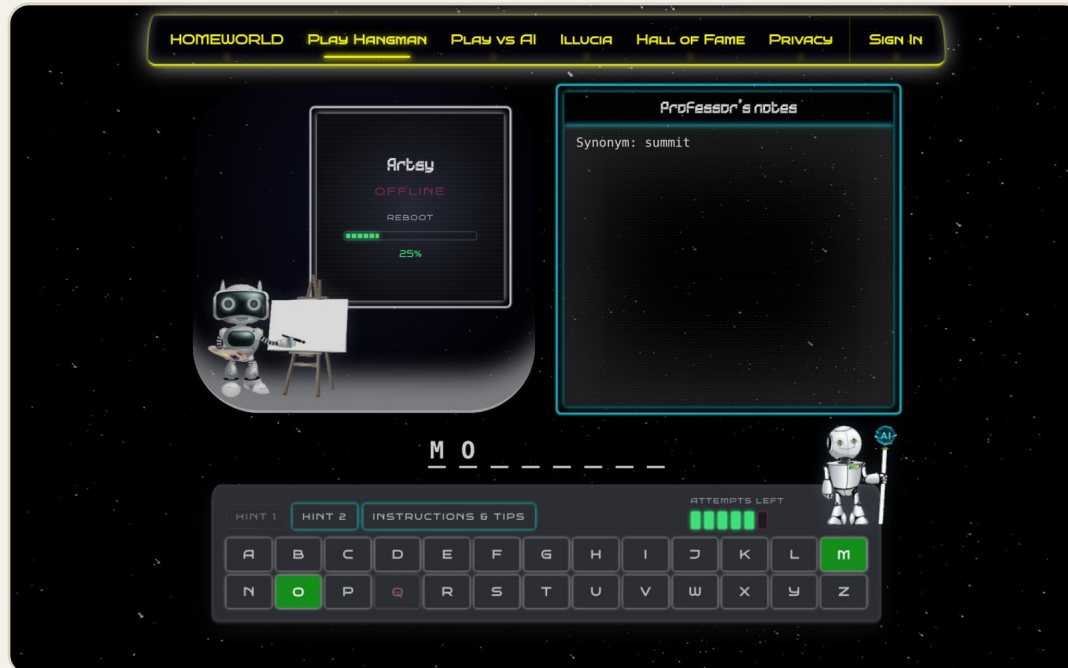
A modern stack

Vite, Hono and D1 replace Create React App, Express and MongoDB.

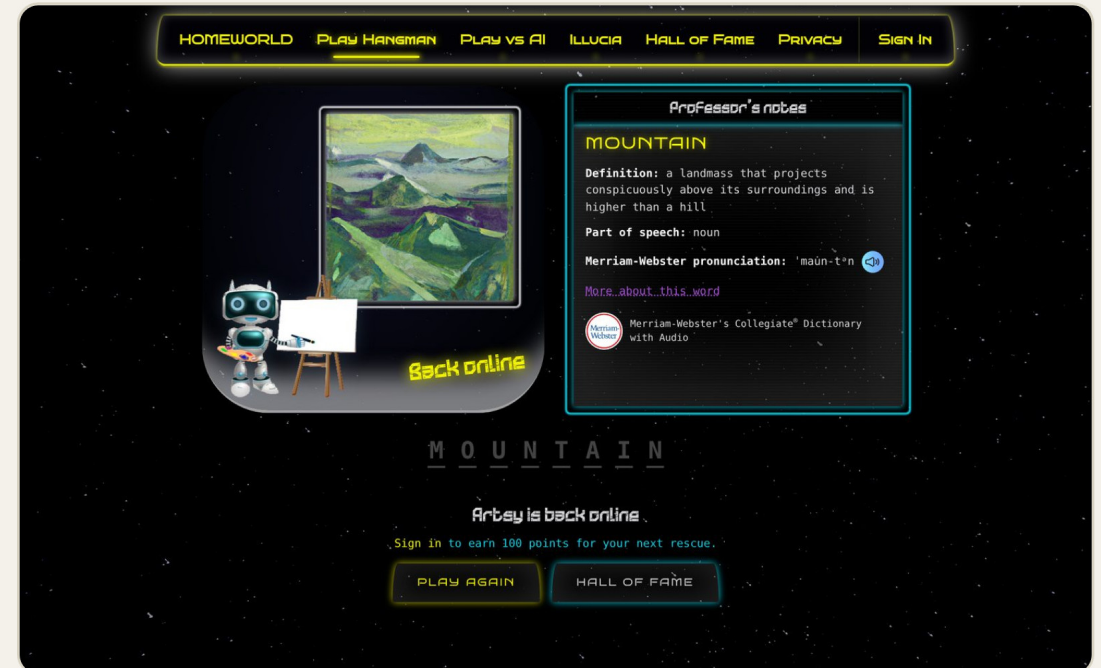
A \$0 running cost

Free-tier allowances are ceilings, never a reason to upgrade.

Save a robot by guessing the word

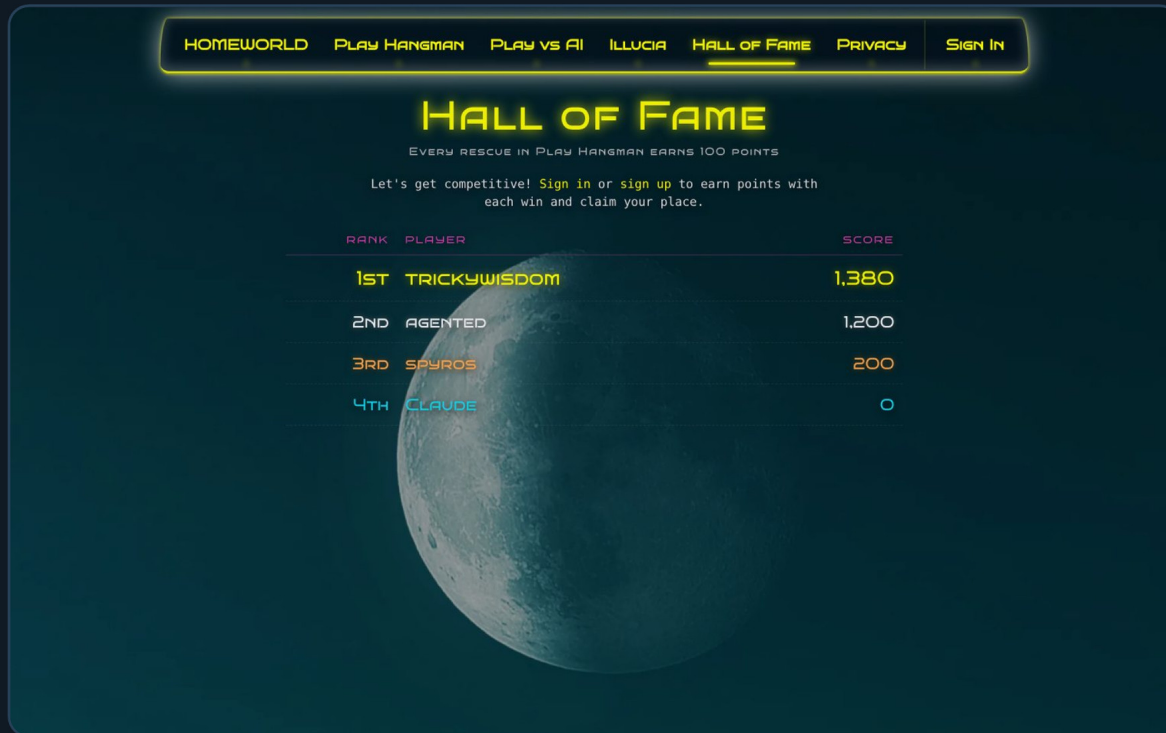


The game. Each right letter reboots Artsy. Each wrong one costs an attempt.

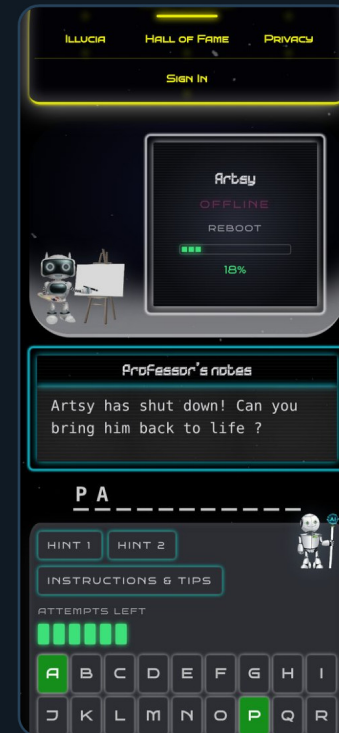


The reward. A win shows a rescued painting and the dictionary entry, with attribution.

A public leaderboard, on any screen



Hall of Fame. Real accounts, real scores.



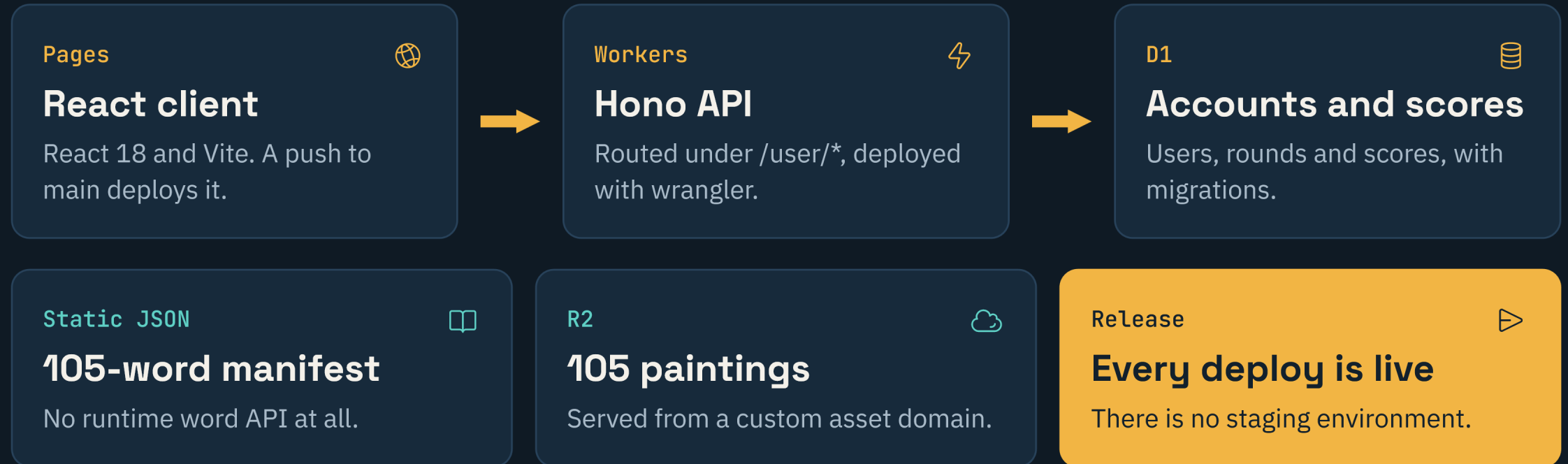
On a phone.

Worth knowing

Guests play for free. Signed-in players earn 100 points per rescue.

The phone layout got its own pass, confirmed on a real device.

All on Cloudflare's free tier



What the \$0 constraint looks like

\$0

Monthly running cost. Free allowances are ceilings, never targets.

105

Words in the game, each with a rescued painting served from R2.

135,897

Words in the Illucia vocabulary, 4 to 15 letters, profanity-filtered.

1.96 ms

Median Worker CPU at launch on the Free plan (3.56 ms at P99, small sample).

Assembled offline, never generated at runtime

Step 1



Merriam-Webster

Collegiate Dictionary only: definition, pronunciation, audio and a real, attributed example.

Step 2



Human overrides

76 explicit entry choices settle ambiguity. No hidden ranking decides.

Step 3



Reviewed enrichment

Hints and explanations drafted with an LLM, human-reviewed, committed as static data.

Step 4



Assemble and validate

A deterministic assembler builds the manifest. A validator rejects hints that leak the answer.

The raw password never leaves the browser

In the browser



PBKDF2-HMAC-SHA-256, 600,000 iterations, a random salt per account.

In the Worker



Only an HMAC verifier under a secret pepper. The derived credential is never stored or logged.

Session



A jose JWT in an HttpOnly cookie, expiring after 24 hours.

The reasoning

Bcrypt hit the Worker's CPU limit, so the work moved to the browser.

The work factor only ever rises, through a new kdf_version. It is never lowered to fit a limit.

A leaderboard that states its limits

Ticket



Server picks the word

The Worker issues an account-bound round ticket. Play itself stays local.

Claim



Replay, then award

A win sends the guesses. The Worker replays them and awards 100 points once.

Floor



Five-second minimum

No earlier claim is accepted, so each account tops out at 12 awards a minute.

Answers are public by design, so valid scores do not prove honest play. The Hall of Fame is not a trusted ranking.

Measuring her broke an assumption

Win rate against common words	Apprentice	Scholar	Master
Before: 4 letters, unweighted	56%	46%	42%
After: 4 to 6 letters, weighted	72.9%	72.3%	72.5%

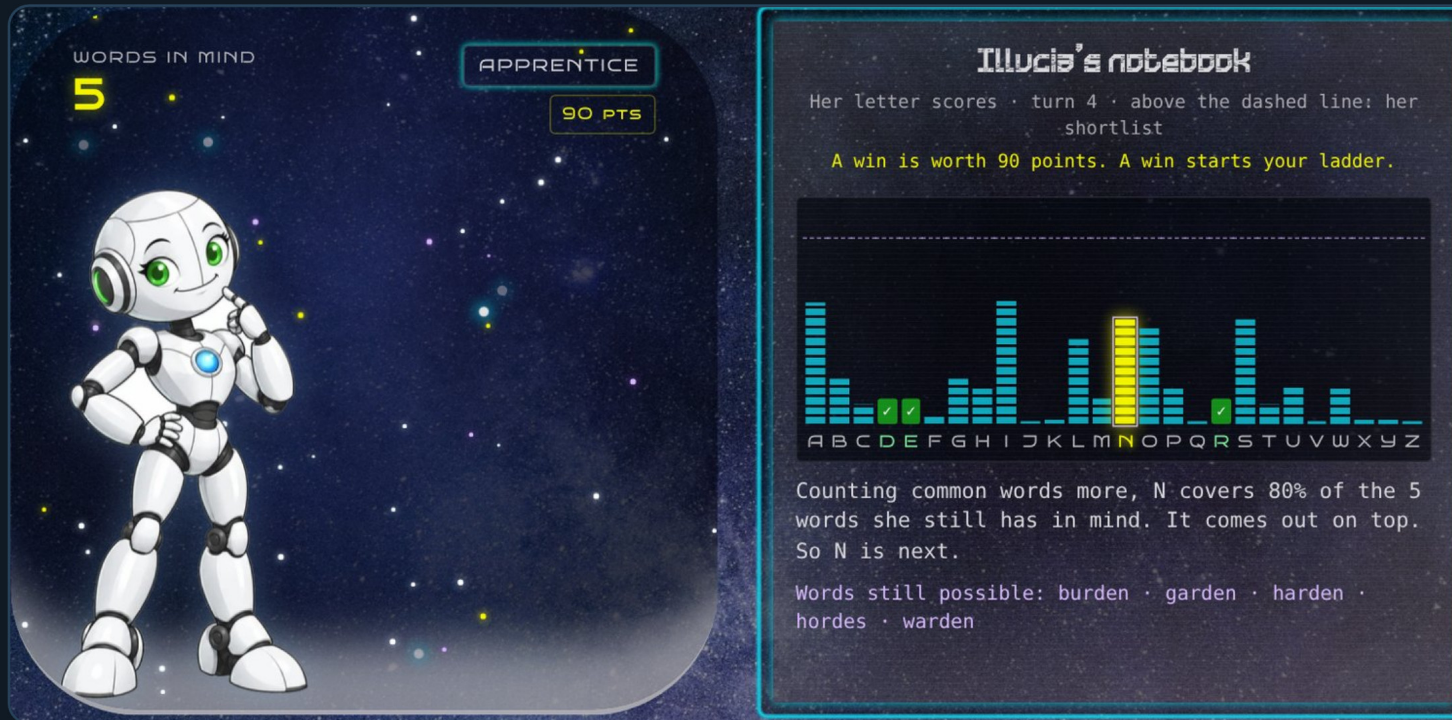
What I found

A bigger vocabulary holds more short look-alikes, so the cleverer tiers played worse on common words.

What I changed

Candidates are weighted by how common they are. A gate now checks tier order: 0 of 171 comparisons fail.

She explains each guess, and it is true



The chart

One bar per letter. Her shortlist sits above the dashed line.

The line

Written from her real decision record, not from a template.

Why

A line that describes something she did not do is a bug.

The Observatory. A live duel, Apprentice level, turn 4.

Chat plans, Code executes, tests decide

Plan



In Claude chat

A Project holds the instructions, a living state.md and a build diary. Decisions are made here.

Execute



In Claude Code

One scoped task per session. CLAUDE.md carries the constraints, and the opening prompt is the handoff.

Verify



Before trusting

Real diffs and real test output before each commit. A test counts only once I have seen it fail.

One concern per change. One commit per change. Flag every adjacent fix, never bundle it silently.

Built, tested and live are three different claims

Area	State
Accounts and Hall of Fame	Live. Signup, login and a saved win were verified in a browser.
Round-ticket scoring	Live. Production API checks passed; a production browser check was blocked.
Illucia duel and Observatory	Live for signed-in players. No authenticated production check of a points claim is recorded.
Account deletion, round expiry	Deployed. No production check of deletion or a retention run is recorded.
Experimental AI mode	Page tested against a mocked route only. The Worker route is deployed.

Still redesigning, still inventing

- Illucia v2: player memory, WordNet questions and an experimental AI mode, still to be checked live
- Close the open production checks: scoring in a browser, account deletion, Illucia claims
- Self-host the Merriam-Webster audio, if the licence allows it
- Clear the open dependency advisories

Play it

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Read the code

github.com/spyrostrimis/hangman